

CRT Example: Chinese Remainder Theorem

SEC 595: Secure Computing – Lecture 1 Breakdown

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1. Introduction and Ring Isomorphism

In Section 6 of the course notes, the Chinese Remainder Theorem (CRT) is presented not merely as a step-by-step algorithm to solve linear congruences, but as a formal ring isomorphism:

$$\psi : \mathbb{Z}_M \cong \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_k}$$

where $m_1, \dots, m_k$ are pairwise coprime moduli, and $M = m_1 m_2 \cdots m_k$.

2. Breakdown of Example 11 (Lecture 1, Section 6.2)

Example 11 demonstrates how to use idempotents (e_i) to compute the inverse map ψ^{-1} . This allows us to take a target tuple of remainders (a_1, a_2) and uniquely reconstruct the single integer $x \in \mathbb{Z}_M$.

Problem Setup

- **Sub-moduli:** $m_1 = 5$ and $m_2 = 13$
- **Combined modulus:** $M = m_1 \times m_2 = 5 \times 13 = 65$
- **Target tuple to reconstruct:** $(a_1, a_2) = (3, 10)$

Construction of Idempotents (e_1 and e_2)

An idempotent e_i acts as a mathematical "switch" that equals 1 (mod m_i) and 0 (mod m_j) for all $j \neq i$. The formula for the i -th idempotent is:

$$e_i = M_i \cdot y_i \pmod{M}$$

where $M_i = M/m_i$ and $y_i = M_i^{-1} \pmod{m_i}$.

1. Computing e_1 (for $m_1 = 5$):

- $M_1 = 65/5 = 13$.
- Reduce M_1 modulo m_1 : $13 \equiv 3 \pmod{5}$.
- Find $y_1 = 3^{-1} \pmod{5}$. Since $3 \times 2 = 6 \equiv 1 \pmod{5}$, we have $y_1 = 2$.
- Compute idempotent: $e_1 = M_1 \cdot y_1 = 13 \times 2 = 26$.

2. Computing e_2 (for $m_2 = 13$):

- $M_2 = 65/13 = 5$.
- Find $y_2 = 5^{-1} \pmod{13}$. Since $5 \times 8 = 40 \equiv 1 \pmod{13}$, we have $y_2 = 8$.
- Compute idempotent: $e_2 = M_2 \cdot y_2 = 5 \times 8 = 40$.

Verifying Idempotent Properties and Partition of Unity

The example verifies that $e_1 = 26$ and $e_2 = 40$ behave correctly:

$$\text{Check } e_1 = 26 : \quad 26 \equiv 1 \pmod{5}, \quad 26 \equiv 0 \pmod{13}$$

$$\text{Check } e_2 = 40 : \quad 40 \equiv 0 \pmod{5}, \quad 40 \equiv 1 \pmod{13}$$

Partition of Unity Check:

$$e_1 + e_2 = 26 + 40 = 66 \equiv 1 \pmod{65}$$

Reconstructing the Solution ($x = \psi^{-1}(3, 10)$)

To map back from $(3, 10)$ to $\mathbb{Z}_{65}$, we evaluate the linear combination:

$$\begin{aligned} x &\equiv a_1 e_1 + a_2 e_2 \pmod{M} \\ &\equiv (3 \cdot 26) + (10 \cdot 40) \pmod{65} \\ &\equiv 78 + 400 \pmod{65} \\ &\equiv 478 \pmod{65} \end{aligned}$$

Dividing 478 by 65 yields:

$$478 = (7 \times 65) + 23 \Rightarrow x = 23$$

Final Verification

Checking that $x = 23$ correctly yields the target tuple $(3, 10)$:

$$23 \pmod{5} = 3 \quad \text{and} \quad 23 \pmod{13} = 10$$

The ring isomorphism successfully and uniquely recovers the integer $23 \in \mathbb{Z}_{65}$.